

Word Problems & Reasoning

ANSWER KEY



Number and digit reasoning

- 1 A number decreased by 9 equals 34. Find the number.

$$x - 9 = 34 \Rightarrow x = 43.$$

- 2 Three times a number, minus 7, equals 32. Find the number.

$$3x - 7 = 32 \Rightarrow 3x = 39 \Rightarrow x = 13.$$

- 3 The sum of two numbers is 58. One number is 12 more than the other. Find both numbers.

Let the smaller be x : $x + (x + 12) = 58 \Rightarrow 2x = 46 \Rightarrow x = 23$. The numbers are 23 and 35.

- 4 A two-digit number has digit sum 11. Reversing its digits gives a number 45 more than the original. Find the original number.

Let the digits be t (tens) and u (units): $t + u = 11$ and $(10u + t) - (10t + u) = 45 \Rightarrow u - t = 5$. Solving together: $u = 8, t = 3$. The original number is 38.

Age problems

- 5 A father is 4 times as old as his daughter. In 5 years, he will be 3 times as old as she will be. Find their current ages.

Let the daughter be x , father $4x$: $4x + 5 = 3(x + 5) \Rightarrow 4x + 5 = 3x + 15 \Rightarrow x = 10$. Daughter is 10, father is 40.

- 6 Sara is twice as old as her brother. In 6 years, the sum of their ages will be 42. Find their current ages.

Let the brother be x , Sara $2x$: $(x + 6) + (2x + 6) = 42 \Rightarrow 3x = 24 \Rightarrow x = 10$. Brother is 10, Sara is 20.

- 7 Ahmed is 5 years older than Salem. Seven years ago, Ahmed was twice as old as Salem. Find their current ages.

Let Salem be x , Ahmed $x + 5$: $(x + 5 - 7) = 2(x - 7) \Rightarrow x - 2 = 2x - 14 \Rightarrow x = 12$. Salem is 12, Ahmed is 17.

- 8 The sum of a mother's and daughter's ages is 50. The mother is 20 years older than the daughter. Find both ages.

Let the daughter be d : $d + (d + 20) = 50 \Rightarrow 2d = 30 \Rightarrow d = 15$. Daughter is 15, mother is 35.

Distance, rate and time

- 9 A train travels 180 km in 2 hours. At the same speed, how far does it travel in 5 hours?

$$\text{Speed} = 180 \div 2 = 90 \text{ km/h. Distance} = 90 \times 5 = 450 \text{ km.}$$

- 10 Two cities are 350 km apart. A car leaves one city for the other at 70 km/h. How long does the trip take?

$$\text{Time} = 350 \div 70 = 5 \text{ hours.}$$

- 11 Two cyclists start at the same point and ride in opposite directions, at 15 km/h and 20 km/h. After how many hours will they be 105 km apart?

$$\text{Combined speed} = 15 + 20 = 35 \text{ km/h. Time} = 105 \div 35 = 3 \text{ hours.}$$

- 12 A plane flies against a headwind at 480 km/h and covers 1,440 km. How many hours does the flight take?

$$\text{Time} = 1,440 \div 480 = 3 \text{ hours.}$$

Work-rate problems

- 13 Worker A can paint a room in 6 hours; Worker B can paint the same room in 3 hours. Working together, how long will it take them?

$$\text{Combined rate} = \frac{1}{6} + \frac{1}{3} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6} = \frac{1}{2} \text{ room/hour. Time} = 2 \text{ hours.}$$

- 14 A pipe fills a tank in 10 hours; a second pipe fills it in 15 hours. Working together, how long to fill the tank?

$$\text{Combined rate} = \frac{1}{10} + \frac{1}{15} = \frac{3}{30} + \frac{2}{30} = \frac{5}{30} = \frac{1}{6}. \text{ Time} = 6 \text{ hours.}$$

- 15 6 workers can complete a job in 8 days. How many days would 4 workers take, at the same rate?

$$\text{Total work} = 6 \times 8 = 48 \text{ worker-days. With 4 workers: } 48 \div 4 = 12 \text{ days.}$$

- 16 A pipe fills a tank in 12 hours, but a leak can empty a full tank in 24 hours. If both act at once, how long to fill the tank?

$$\text{Net rate} = \frac{1}{12} - \frac{1}{24} = \frac{2}{24} - \frac{1}{24} = \frac{1}{24}. \text{ Time} = 24 \text{ hours.}$$

Mixture and money problems

- 17 A jeweler mixes gold worth 200 riyals/gram with silver worth 50 riyals/gram to make 10 grams of a mixture worth 110 riyals/gram. How many grams of gold are used?

$$\text{Let } g \text{ be grams of gold: } 200g + 50(10 - g) = 110 \times 10 \Rightarrow 150g = 600 \Rightarrow g = 4 \text{ grams.}$$

- 18** A shopkeeper has 15 riyals in 1-riyal and 2-riyal coins, 10 coins in total. How many of each coin does she have?

Let x be 1-riyal coins, y be 2-riyal coins: $x + y = 10$, $x + 2y = 15$. Subtracting: $y = 5$, $x = 5$.

- 19** A solution is 40% acid. How many liters of pure water must be added to 20 liters of this solution to dilute it to 25% acid?

Acid amount stays $0.4 \times 20 = 8$ liters. $\frac{8}{20 + w} = 0.25 \Rightarrow 20 + w = 32 \Rightarrow w = 12$ liters.

- 20** Event tickets cost 30 riyals for adults and 18 riyals for children. 150 tickets were sold for 3,780 riyals in total. How many adult tickets were sold?

Let a be adults, so $150 - a$ children: $30a + 18(150 - a) = 3,780 \Rightarrow 12a = 1,080 \Rightarrow a = 90$ adult tickets.

Logical deduction

- 21** Ali, Basim and Khalid have 36 books together. Ali has twice as many as Basim, and Khalid has 6 more than Ali. Find how many books each has.

Let Basim = x , Ali = $2x$, Khalid = $2x + 6$: $x + 2x + 2x + 6 = 36 \Rightarrow 5x = 30 \Rightarrow x = 6$. Basim has 6, Ali has 12, Khalid has 18.

- 22** A number between 50 and 70 is divisible by both 3 and 4, but not by 8. Find the number.

Multiples of 12 between 50 and 70: only 60. Is 60 divisible by 8? $60 \div 8 = 7.5$, no. The number is 60.

- 23** In a class, every student plays football or basketball (or both). 25 play football, 18 play basketball, and 10 play both. How many students are in the class?

By inclusion-exclusion: $25 + 18 - 10 = 33$ students.

- 24** Four consecutive even integers sum to 92. Find the largest one.

Let the integers be $n, n + 2, n + 4, n + 6$: $4n + 12 = 92 \Rightarrow n = 20$. The largest is 26.

Working backward

- 25** After spending half his money on a book, then 20 riyals on lunch, Yousef has 30 riyals left. How much did he start with?

Let the start be x : $\frac{x}{2} - 20 = 30 \Rightarrow \frac{x}{2} = 50 \Rightarrow x = 100$ riyals.

- 26** A number is tripled, then 8 is subtracted, giving 25. Find the original number.

$3x - 8 = 25 \Rightarrow 3x = 33 \Rightarrow x = 11$.

- 27 After a 15% pay cut, an employee's salary is 3,400 riyals. Find the salary before the cut.

$$\text{Original} \times 0.85 = 3,400 \Rightarrow \text{original} = 3,400 \div 0.85 = 4,000 \text{ riyals.}$$

- 28 A water tank is $\frac{1}{4}$ full. After adding 60 liters, it becomes $\frac{5}{8}$ full. Find the tank's total capacity.

$$60 = C\left(\frac{5}{8} - \frac{1}{4}\right) = C\left(\frac{3}{8}\right) \Rightarrow C = 60 \times \frac{8}{3} = 160 \text{ liters.}$$

Mixed word problems

- 29 A rectangular field is twice as long as it is wide, and its perimeter is 90 m. Find its length and width.

$$\text{Let width} = w, \text{ length} = 2w: 2(w + 2w) = 90 \Rightarrow 6w = 90 \Rightarrow w = 15 \text{ m. Length} = 30 \text{ m.}$$

- 30 Layla buys 3 notebooks and 2 pens for 29 riyals. Fahad buys 1 notebook and 4 pens for 23 riyals, at the same prices. Find the price of one notebook and one pen.

$$\text{Let } n \text{ be the notebook price, } p \text{ the pen price: } 3n + 2p = 29, n + 4p = 23 \Rightarrow n = 23 - 4p. \text{ Substituting: } 3(23 - 4p) + 2p = 29 \Rightarrow -10p = -40 \Rightarrow p = 4, n = 7. \text{ Notebook: 7 riyals, pen: 4 riyals.}$$

- 31 A car rental costs 80 riyals per day plus 0.50 riyals per km driven. A 3-day rental cost 435 riyals in total. How many km were driven?

$$80 \times 3 + 0.5k = 435 \Rightarrow 240 + 0.5k = 435 \Rightarrow 0.5k = 195 \Rightarrow k = 390 \text{ km.}$$

- 32 The sum of three consecutive odd integers is 63. Find the integers.

$$\text{Let the smallest be } n: n + (n + 2) + (n + 4) = 63 \Rightarrow 3n + 6 = 63 \Rightarrow n = 19. \text{ The integers are 19, 21, 23.}$$

- 33 A shop sells notebooks for 6 riyals each, with a "buy 3, get 1 free" offer. Layla needs 20 notebooks. How much will she pay?

$$\text{Every group of 4 notebooks costs for 3: } 20 \div 4 = 5 \text{ groups, each costing } 3 \times 6 = 18 \text{ riyals. Total} = 5 \times 18 = 90 \text{ riyals.}$$

More age and number problems

- 34 The sum of a number and its square is 42. Find the positive value of the number.

$$x + x^2 = 42 \Rightarrow x^2 + x - 42 = 0 \Rightarrow (x + 7)(x - 6) = 0. \text{ The positive value is } x = 6.$$

- 35 A number's triple exceeds its double by 15. Find the number.

$$3x = 2x + 15 \Rightarrow x = 15.$$

- 36 In 10 years, Noura will be 3 times as old as she was 2 years ago. Find her current age.

Let her current age be x : $x + 10 = 3(x - 2) \Rightarrow x + 10 = 3x - 6 \Rightarrow 16 = 2x \Rightarrow x = 8$.

- 37 The sum of two consecutive multiples of 5 is 75. Find the larger one.

Let the numbers be $5k$ and $5k + 5$: $10k + 5 = 75 \Rightarrow k = 7$. The numbers are 35 and 40; the larger is 40.

More rate and proportion problems

- 38 A photocopier prints 45 pages per minute. How many minutes does it take to print 810 pages?

$810 \div 45 = 18$ minutes.

- 39 A recipe for 6 people uses 450 g of rice. How much rice is needed for 15 people, at the same rate per person?

Rice per person = $450 \div 6 = 75$ g. For 15 people: $15 \times 75 = 1,125$ g.

- 40 A factory produces 1,200 items in 8 hours. At the same rate, how many items would it produce in a 10-hour shift?

Rate = $1,200 \div 8 = 150$ items/hour. In 10 hours: $10 \times 150 = 1,500$ items.

- 41 Three taps together fill a pool in 4 hours. Two of the taps alone would take 6 hours. How long would the third tap alone take to fill the pool?

Combined rate = $\frac{1}{4}$. Two-tap rate = $\frac{1}{6}$. Third tap rate = $\frac{1}{4} - \frac{1}{6} = \frac{3}{12} - \frac{2}{12} = \frac{1}{12}$, so it alone takes 12 hours.

MCQ — Number and digit reasoning

- 42 A number decreased by 12 equals 47. Find the number.

a. 59 ✓

b. 65

c. 35

d. 47

$x - 12 = 47 \Rightarrow x = 59$.

43 Four times a number, minus 9, equals 47. Find the number.

a. 56

b. 14 ✓

c. 11

d. 9.5

$$4x - 9 = 47 \Rightarrow 4x = 56 \Rightarrow x = 14.$$

44 The sum of two numbers is 74. One number is 18 more than the other. Find the larger number.

a. 46 ✓

b. 37

c. 28

d. 56

$$x + (x + 18) = 74 \Rightarrow x = 28; \text{larger} = 46.$$

45 A two-digit number has digit sum 9. Reversing its digits gives a number 27 more than the original. Find the original number.

a. 18

b. 36 ✓

c. 63

d. 45

$$\text{Digits 3, 6: original} = 36, \text{ reversed} = 63 = 36 + 27.$$

46 A number is increased by 25% to give 60. Find the original number.

- a. 75
- b. 40
- c. 45
- d. 48 ✓

$$1.25x = 60 \Rightarrow x = 48.$$

47 A number plus twice itself equals 27. Find the number.

- a. 13.5
- b. 27
- c. 9 ✓
- d. 18

$$x + 2x = 27 \Rightarrow 3x = 27 \Rightarrow x = 9.$$

MCQ — Age problems

48 A father is 5 times as old as his son. In 4 years, he will be 3 times as old as his son will be. Find the father's current age.

- a. 20 ✓
- b. 16
- c. 24
- d. 4

$$5x + 4 = 3(x + 4) \Rightarrow 2x = 8 \Rightarrow x = 4; \text{ father} = 20.$$

49 Huda is twice as old as her sister. In 8 years, the sum of their ages will be 46. Find Huda's current age.

- a. 10
- b. 18
- c. 14
- d. 20 ✓

$$(x + 8) + (2x + 8) = 46 \Rightarrow x = 10; \text{Huda} = 20.$$

50 Khalid is 6 years older than Faisal. Five years ago, Khalid was twice as old as Faisal. Find their current ages.

- a. Faisal = 17, Khalid = 11
- b. Faisal = 11, Khalid = 6
- c. Faisal = 5, Khalid = 11
- d. Faisal = 11, Khalid = 17 ✓

$$(x + 6 - 5) = 2(x - 5) \Rightarrow x + 1 = 2x - 10 \Rightarrow x = 11; \text{Khalid} = 17.$$

51 The sum of a mother's and son's ages is 54. The mother is 24 years older than the son. Find the son's age.

- a. 30
- b. 15 ✓
- c. 27
- d. 39

$$x + (x + 24) = 54 \Rightarrow 2x = 30 \Rightarrow x = 15.$$

52 In 8 years, Mona will be twice as old as she was 4 years ago. Find her current age.

- a. 8
- b. 20
- c. 16 ✓
- d. 12

$$x + 8 = 2(x - 4) \Rightarrow x + 8 = 2x - 8 \Rightarrow x = 16.$$

53 Tariq is currently twice as old as his son. If Tariq is 40, find his son's age.

- a. 18
- b. 20 ✓
- c. 22
- d. 80

$$40 = 2s \Rightarrow s = 20.$$

MCQ — Distance, rate and time

54 A train travels 210 km in 3 hours. At the same speed, how far does it travel in 7 hours?

- a. 490 ✓
- b. 420
- c. 70
- d. 630

$$\text{Speed} = 70 \text{ km/h}; 70 \times 7 = 490 \text{ km}.$$

55 Two cities are 420 km apart. A car leaves one city for the other at 60 km/h. How long does the trip take?

- a. 6 hours
- b. 25,200 hours
- c. 7 hours ✓
- d. 8 hours

$$420 \div 60 = 7 \text{ hours.}$$

56 Two cyclists start at the same point and ride in opposite directions, at 12 km/h and 18 km/h. After 3 hours, how far apart are they?

- a. 54
- b. 90 ✓
- c. 30
- d. 36

$$\text{Combined speed} = 30 \text{ km/h; } 30 \times 3 = 90 \text{ km.}$$

57 A plane flies against a headwind at 400 km/h and covers 1,600 km. How many hours does the flight take?

- a. 4,000
- b. 4 ✓
- c. 5
- d. 400

$$1,600 \div 400 = 4 \text{ hours.}$$

58 A car travels at 90 km/h for 2.5 hours. Find the distance covered.

a. 225 ✓

b. 36

c. 92.5

d. 180

$$90 \times 2.5 = 225 \text{ km.}$$

MCQ — Work-rate problems

59 Worker A can paint a room in 8 hours; Worker B can paint the same room in 4 hours. Working together, how long will it take?

a. 4 hours

b. 12 hours

c. 6 hours

d. $\frac{8}{3}$ hours ✓

$$\frac{1}{8} + \frac{1}{4} = \frac{3}{8} \text{ per hour; time} = \frac{8}{3} \text{ hours.}$$

60 A pipe fills a tank in 12 hours; a second pipe fills it in 6 hours. Working together, how long to fill the tank?

a. 3 hours

b. 18 hours

c. 4 hours ✓

d. 9 hours

$$\frac{1}{12} + \frac{1}{6} = \frac{1}{4} \text{ per hour; time} = 4 \text{ hours.}$$

61 8 workers can complete a job in 6 days. How many days would 12 workers take, at the same rate?

a. 4 ✓

b. 9

c. 3

d. 6

Total work = 48 worker-days; $48 \div 12 = 4$ days.

62 A pipe fills a tank in 10 hours, but a leak can empty a full tank in 20 hours. If both act at once, how long to fill the tank?

a. 20 hours ✓

b. 30 hours

c. 10 hours

d. 15 hours

Net rate = $\frac{1}{10} - \frac{1}{20} = \frac{1}{20}$ per hour; time = 20 hours.

63 Worker C can complete a task in 9 hours. After working alone for 3 hours, what fraction of the task remains?

a. $\frac{2}{3}$ ✓

b. $\frac{1}{3}$

c. $\frac{1}{6}$

d. $\frac{1}{9}$

In 3 hours, $\frac{3}{9} = \frac{1}{3}$ is done; remaining = $\frac{2}{3}$.

MCQ — Mixture and money problems

- 64 A jeweler mixes gold worth 250 riyals/gram with silver worth 40 riyals/gram to make 12 grams of a mixture worth 145 riyals/gram. How many grams of gold are used?

a. 6 ✓

b. 4

c. 8

d. 9

$$250g + 40(12 - g) = 1,740 \Rightarrow 210g = 1,260 \Rightarrow g = 6.$$

- 65 A shopkeeper has 21 riyals in 1-riyal and 2-riyal coins, 15 coins in total. How many 2-riyal coins does he have?

a. 9

b. 15

c. 6 ✓

d. 3

$$x + y = 15, x + 2y = 21 \Rightarrow y = 6.$$

- 66 A solution is 30% acid. How many liters of pure water must be added to 15 liters of this solution to make it 20% acid?

a. 4.5

b. 7.5 ✓

c. 5

d. 10

$$\frac{4.5}{15 + w} = 0.2 \Rightarrow 15 + w = 22.5 \Rightarrow w = 7.5.$$

- 67 Event tickets cost 25 riyals for adults and 15 riyals for children. 200 tickets were sold for 4,300 riyals. How many adult tickets were sold?

- a. 150
- b. 130 ✓
- c. 100
- d. 70

$$25a + 15(200 - a) = 4,300 \Rightarrow 10a = 1,300 \Rightarrow a = 130.$$

- 68 A syrup is 25% sugar. How many kilograms of pure sugar must be added to 8 kg of this syrup to make it 40% sugar?

- a. 3.2
- b. 4
- c. 1
- d. 2 ✓

$$\frac{2 + x}{8 + x} = 0.4 \Rightarrow 2 + x = 3.2 + 0.4x \Rightarrow x = 2.$$

MCQ — Logical deduction

- 69 Omar, Bandar and Nasser have 44 books together. Omar has twice as many as Bandar, and Nasser has 8 more than Bandar. Find how many books Bandar has.

- a. 22
- b. 17
- c. 9 ✓
- d. 18

$$b + 2b + (b + 8) = 44 \Rightarrow 4b = 36 \Rightarrow b = 9.$$

70 A number between 80 and 100 is divisible by both 4 and 6, but not by 8. Find the number.

a. 84 ✓

b. 96

c. 88

d. 90

Multiples of 12 in range: 84, 96. $84 \div 8 = 10.5$ (not divisible); $96 \div 8 = 12$ (excluded).

71 In a class, every student plays chess or checkers (or both). 20 play chess, 15 play checkers, and 8 play both. How many students are in the class?

a. 27 ✓

b. 35

c. 12

d. 43

$20 + 15 - 8 = 27$.

72 Four consecutive odd integers sum to 96. Find the largest one.

a. 30

b. 27 ✓

c. 24

d. 21

$4n + 12 = 96 \Rightarrow n = 21$; largest $= n + 6 = 27$.

73 Three consecutive integers sum to 72. Find the smallest one.

- a. 25
- b. 24
- c. 23 ✓
- d. 22

$$3n + 3 = 72 \Rightarrow n = 23.$$

MCQ — Working backward

74 After spending half his money on a book, then 15 riyals on lunch, Faisal has 25 riyals left. How much did he start with?

- a. 100
- b. 65
- c. 80 ✓
- d. 40

$$\frac{x}{2} - 15 = 25 \Rightarrow \frac{x}{2} = 40 \Rightarrow x = 80.$$

75 A number is doubled, then 6 is subtracted, giving 20. Find the original number.

- a. 16
- b. 7
- c. 26
- d. 13 ✓

$$2x - 6 = 20 \Rightarrow x = 13.$$

76 After a 20% pay cut, an employee's salary is 3,200 riyals. Find the salary before the cut.

a. 4,200

b. 4,000 ✓

c. 3,600

d. 3,840

$$0.8x = 3,200 \Rightarrow x = 4,000.$$

77 A water tank is $\frac{1}{3}$ full. After adding 80 liters, it becomes $\frac{3}{4}$ full. Find the tank's total capacity.

a. 240

b. 192 ✓

c. 120

d. 106.7

$$\frac{C}{3} + 80 = \frac{3C}{4} \Rightarrow 80 = \frac{5C}{12} \Rightarrow C = 192.$$

78 A number is halved, then 10 is added, giving 34. Find the original number.

a. 48 ✓

b. 44

c. 88

d. 24

$$\frac{x}{2} + 10 = 34 \Rightarrow \frac{x}{2} = 24 \Rightarrow x = 48.$$

MCQ — Mixed word problems

79 A rectangular field is twice as long as it is wide, and its perimeter is 108 m. Find its length and width.

- a. Length = 36, width = 36
- b. Length = 18, width = 36
- c. Length = 54, width = 27
- d. Length = 36, width = 18 ✓

$$2(w + 2w) = 108 \Rightarrow w = 18; \text{length} = 36.$$

80 Sarah buys 4 notebooks and 3 pens for 38 riyals. Khalid buys 2 notebooks and 5 pens for 40 riyals, at the same prices. Find the price of one notebook.

- a. 6
- b. 8
- c. 5 ✓
- d. 4

Solving the system: notebook = 5 riyals, pen = 6 riyals.

81 A car rental costs 100 riyals per day plus 0.50 riyals per km driven. A 4-day rental cost 500 riyals. How many km were driven?

- a. 250
- b. 1,000
- c. 200 ✓
- d. 100

$$100(4) = 400; \text{remaining} = 100; \text{km} = 100 \div 0.5 = 200.$$

82 The sum of three consecutive even integers is 78. Find the largest one.

a. 28 ✓

b. 26

c. 30

d. 24

$$3n + 6 = 78 \Rightarrow n = 24; \text{ largest} = n + 4 = 28.$$

83 A shop sells pens for 4 riyals each, with a "buy 4, get 1 free" offer. Karim needs 25 pens. How much will he pay?

a. 100

b. 64

c. 80 ✓

d. 96

$$25 \text{ pens} = 5 \text{ groups of } 5 \text{ (pay for 4, get 1 free): } 5 \times 4 \times 4 = 80 \text{ riyals.}$$

84 A cyclist travels 60 km at a certain speed, then returns the same 60 km at half that speed, taking 9 hours total. Find the original speed.

a. 15

b. 10

c. 40

d. 20 ✓

$$\frac{60}{v} + \frac{120}{v} = 9 \Rightarrow \frac{180}{v} = 9 \Rightarrow v = 20 \text{ km/h.}$$

85 Two numbers have a ratio of 3 : 5 and a sum of 96. Find the larger number.

a. 60 ✓

b. 48

c. 36

d. 72

8 parts = 96, each part = 12; larger = $5 \times 12 = 60$.

MCQ — More rate and proportion problems

86 A photocopier prints 60 pages per minute. How many minutes does it take to print 900 pages?

a. 18

b. 12

c. 54,000

d. 15 ✓

$900 \div 60 = 15$ minutes.

87 A recipe for 8 people uses 600 g of rice. How much rice is needed for 20 people, at the same rate per person?

a. 750

b. 1,500 ✓

c. 1,800

d. 1,200

Rate = 75 g/person; $20 \times 75 = 1,500$ g.

- 88 A factory produces 1,500 items in 6 hours. At the same rate, how many items would it produce in a 9-hour shift?
- a. 1,875
 - b. 2,700
 - c. 2,250 ✓
 - d. 2,000

Rate = 250/hr; $9 \times 250 = 2,250$.

- 89 Three taps together fill a pool in 5 hours. Two of the taps alone would take 8 hours. How long would the third tap alone take?
- a. 3 hours
 - b. 13 hours
 - c. $\frac{40}{3}$ hours ✓
 - d. 40 hours

Third tap's rate = $\frac{1}{5} - \frac{1}{8} = \frac{3}{40}$ per hour; time = $\frac{40}{3}$ hours.

- 90 A car uses 8 liters of fuel to travel 96 km. At the same rate, how many liters are needed to travel 156 km?
- a. 13 ✓
 - b. 14.6
 - c. 19.5
 - d. 12

Rate = 12 km/liter; $156 \div 12 = 13$ liters.

91 A map has a scale of 1 : 50,000. If two towns are 6.4 cm apart on the map, find the actual distance in km.

a. 3.2 ✓

b. 0.32

c. 320

d. 32

$$6.4 \times 50,000 = 320,000 \text{ cm} = 3,200 \text{ m} = 3.2 \text{ km}.$$

Multiple choice — easy

92 A number decreased by 15 equals 26. Find the number.

a. 41 ✓

b. 39

c. 46

d. 43

$$x - 15 = 26 \Rightarrow x = 41.$$

93 A number decreased by 29 equals -2. Find the number.

a. 25

b. 29

c. 27 ✓

d. 32

$$x - 29 = -2 \Rightarrow x = 27.$$

94 A number decreased by 3 equals 59. Find the number.

- a. 60
- b. 64
- c. 62 ✓
- d. 67

$$x - 3 = 59 \Rightarrow x = 62.$$

95 A number decreased by 37 equals 8. Find the number.

- a. 43
- b. 45 ✓
- c. 47
- d. 50

$$x - 37 = 8 \Rightarrow x = 45.$$

96 A number decreased by 13 equals 66. Find the number.

- a. 79 ✓
- b. 77
- c. 84
- d. 81

$$x - 13 = 66 \Rightarrow x = 79.$$

97 Three times a number, minus 5, equals 73. Find the number.

- a. 28
- b. 26 ✓
- c. 24
- d. 31

$$3x - 5 = 73 \Rightarrow 3x = 78 \Rightarrow x = 26.$$

98 Four times a number, minus 17, equals 31. Find the number.

a. 10

b. 14

c. 12 ✓

d. 17

$$4x - 17 = 31 \Rightarrow 4x = 48 \Rightarrow x = 12.$$

99 A number decreased by 35 equals -1. Find the number.

a. 34 ✓

b. 36

c. 39

d. 32

$$x - 35 = -1 \Rightarrow x = 34.$$

100 A number decreased by 38 equals -19. Find the number.

a. 21

b. 19 ✓

c. 17

d. 24

$$x - 38 = -19 \Rightarrow x = 19.$$

101 Four times a number, minus 9, equals 59. Find the number.

- a. 15
- b. 19
- c. 22
- d. 17 ✓

$$4x - 9 = 59 \Rightarrow 4x = 68 \Rightarrow x = 17.$$

102 Two times a number, minus 11, equals 21. Find the number.

- a. 21
- b. 18
- c. 16 ✓
- d. 14

$$2x - 11 = 21 \Rightarrow 2x = 32 \Rightarrow x = 16.$$

103 Two times a number, minus 17, equals 51. Find the number.

- a. 39
- b. 32
- c. 34 ✓
- d. 36

$$2x - 17 = 51 \Rightarrow 2x = 68 \Rightarrow x = 34.$$

104 Three times a number, minus 13, equals 17. Find the number.

- a. 15
- b. 8
- c. 12
- d. 10 ✓

$$3x - 13 = 17 \Rightarrow 3x = 30 \Rightarrow x = 10.$$

105 A number decreased by 13 equals 25. Find the number.

- a. 36
- b. 43
- c. 40
- d. 38 ✓

$$x - 13 = 25 \Rightarrow x = 38.$$

106 Three times a number, minus 12, equals 33. Find the number.

- a. 13
- b. 17
- c. 20
- d. 15 ✓

$$3x - 12 = 33 \Rightarrow 3x = 45 \Rightarrow x = 15.$$

Multiple choice — medium

107 A train travels 535 km in 5 hours. At the same speed, how far does it travel in 2 hours?

- a. 219
- b. 212
- c. 214 ✓
- d. 216

$$\text{Speed} = 535 \div 5 = 107 \text{ km/h. Distance} = 107 \times 2 = 214 \text{ km.}$$

108 A train travels 200 km in 2 hours. At the same speed, how far does it travel in 5 hours?

- a. 505
- b. 500 ✓
- c. 502
- d. 498

Speed = $200 \div 2 = 100$ km/h. Distance = $100 \times 5 = 500$ km.

109 A train travels 210 km in 2 hours. At the same speed, how far does it travel in 6 hours?

- a. 635
- b. 632
- c. 628
- d. 630 ✓

Speed = $210 \div 2 = 105$ km/h. Distance = $105 \times 6 = 630$ km.

110 Two cities are 496 km apart. A car leaves one city for the other at 62 km/h. How long does the trip take?

- a. 6
- b. 8 ✓
- c. 13
- d. 10

Time = $496 \div 62 = 8$ hours.

111 Two cities are 294 km apart. A car leaves one city for the other at 49 km/h. How long does the trip take?

- a. 4
- b. 8
- c. 11
- d. 6 ✓

Time = $294 \div 49 = 6$ hours.

112 Two cities are 488 km apart. A car leaves one city for the other at 61 km/h. How long does the trip take?

- a. 6
- b. 13
- c. 8 ✓
- d. 10

Time = $488 \div 61 = 8$ hours.

113 Two cities are 328 km apart. A car leaves one city for the other at 41 km/h. How long does the trip take?

- a. 8 ✓
- b. 6
- c. 10
- d. 13

Time = $328 \div 41 = 8$ hours.

114 A train travels 141 km in 3 hours. At the same speed, how far does it travel in 2 hours?

- a. 94 ✓
- b. 99
- c. 96
- d. 92

Speed = $141 \div 3 = 47$ km/h. Distance = $47 \times 2 = 94$ km.

115 A train travels 202 km in 2 hours. At the same speed, how far does it travel in 5 hours?

- a. 510
- b. 503
- c. 507
- d. 505 ✓

Speed = $202 \div 2 = 101$ km/h. Distance = $101 \times 5 = 505$ km.

116 Two cities are 132 km apart. A car leaves one city for the other at 33 km/h. How long does the trip take?

- a. 2
- b. 4 ✓
- c. 9
- d. 6

Time = $132 \div 33 = 4$ hours.

117 A train travels 606 km in 6 hours. At the same speed, how far does it travel in 2 hours?

- a. 202 ✓
- b. 204
- c. 207
- d. 200

Speed = $606 \div 6 = 101$ km/h. Distance = $101 \times 2 = 202$ km.

118 Two cities are 189 km apart. A car leaves one city for the other at 63 km/h. How long does the trip take?

- a. 5
- b. 3 ✓
- c. 8
- d. 1

Time = $189 \div 63 = 3$ hours.

119 A train travels 540 km in 6 hours. At the same speed, how far does it travel in 8 hours?

a. 720 ✓

b. 725

c. 718

d. 722

Speed = $540 \div 6 = 90$ km/h. Distance = $90 \times 8 = 720$ km.

120 Two cities are 372 km apart. A car leaves one city for the other at 93 km/h. How long does the trip take?

a. 6

b. 4 ✓

c. 9

d. 2

Time = $372 \div 93 = 4$ hours.

121 A train travels 120 km in 2 hours. At the same speed, how far does it travel in 8 hours?

a. 480 ✓

b. 478

c. 482

d. 485

Speed = $120 \div 2 = 60$ km/h. Distance = $60 \times 8 = 480$ km.

Multiple choice — hard

122 Sara is 2 times as old as her younger sibling. In 3 years, the sum of their ages will be 21. Find their current ages.

a. 5, 14

b. 5, 10 ✓

c. 10, 5

d. 7, 10

Let the sibling be x , Sara $2x$: $(x + 3) + (2x + 3) = 21 \Rightarrow 3x + 6 = 21 \Rightarrow 3x = 15 \Rightarrow x = 5$. Sibling is 5, Sara is 10.

123 Yousef is 5 times as old as his younger sibling. In 6 years, the sum of their ages will be 60. Find their current ages.

a. 8, 40 ✓

b. 8, 44

c. 10, 40

d. 40, 8

Let the sibling be x , Yousef $5x$: $(x + 6) + (5x + 6) = 60 \Rightarrow 6x + 12 = 60 \Rightarrow 6x = 48 \Rightarrow x = 8$. Sibling is 8, Yousef is 40.

124 Layla is 3 times as old as her younger sibling. In 7 years, the sum of their ages will be 54. Find their current ages.

a. 30, 10

b. 10, 34

c. 10, 30 ✓

d. 12, 30

Let the sibling be x , Layla $3x$: $(x + 7) + (3x + 7) = 54 \Rightarrow 4x + 14 = 54 \Rightarrow 4x = 40 \Rightarrow x = 10$. Sibling is 10, Layla is 30.

- 125** Omar is 3 times as old as his younger sibling. In 3 years, the sum of their ages will be 34. Find their current ages.

a. 7, 21 ✓

b. 21, 7

c. 9, 21

d. 7, 25

Let the sibling be x , Omar $3x$: $(x + 3) + (3x + 3) = 34 \Rightarrow 4x + 6 = 34 \Rightarrow 4x = 28 \Rightarrow x = 7$. Sibling is 7, Omar is 21.

- 126** Huda is 2 times as old as her younger sibling. In 5 years, the sum of their ages will be 28. Find their current ages.

a. 6, 16

b. 8, 12

c. 6, 12 ✓

d. 12, 6

Let the sibling be x , Huda $2x$: $(x + 5) + (2x + 5) = 28 \Rightarrow 3x + 10 = 28 \Rightarrow 3x = 18 \Rightarrow x = 6$. Sibling is 6, Huda is 12.

- 127** Worker A can complete a job in 12 hours; Worker B can complete the same job in 4 hours. Working together, how long will it take them?

a. 8

b. 3 ✓

c. 2

d. 4

Combined rate $= \frac{1}{12} + \frac{1}{4} = \frac{1}{12} + \frac{3}{12} = \frac{4}{12}$ job/hour. Time = 3 hours.

128 Worker A can complete a job in 12 hours; Worker B can complete the same job in 6 hours. Working together, how long will it take them?

a. 9

b. 4 ✓

c. 3

d. 5

$$\text{Combined rate} = \frac{1}{12} + \frac{1}{6} = \frac{1}{12} + \frac{2}{12} = \frac{3}{12} \text{ job/hour. Time} = 4 \text{ hours.}$$

129 Worker A can complete a job in 40 hours; Worker B can complete the same job in 10 hours. Working together, how long will it take them?

a. 8 ✓

b. 9

c. 25

d. 7

$$\text{Combined rate} = \frac{1}{40} + \frac{1}{10} = \frac{1}{40} + \frac{4}{40} = \frac{5}{40} \text{ job/hour. Time} = 8 \text{ hours.}$$

130 Worker A can complete a job in 56 hours; Worker B can complete the same job in 8 hours. Working together, how long will it take them?

a. 6

b. 8

c. 32

d. 7 ✓

$$\text{Combined rate} = \frac{1}{56} + \frac{1}{8} = \frac{1}{56} + \frac{7}{56} = \frac{8}{56} \text{ job/hour. Time} = 7 \text{ hours.}$$

131 Worker A can complete a job in 8 hours; Worker B can complete the same job in 24 hours. Working together, how long will it take them?

a. 7

b. 6 ✓

c. 16

d. 5

Combined rate = $\frac{1}{8} + \frac{1}{24} = \frac{3}{24} + \frac{1}{24} = \frac{4}{24}$ job/hour. Time = 6 hours.

132 After spending half his money on a book, then 25 riyals on lunch, Yousef has 19 riyals left. How much did he start with?

a. 19

b. 78

c. 88 ✓

d. 98

Let the start be x : $\frac{x}{2} - 25 = 19 \Rightarrow \frac{x}{2} = 44 \Rightarrow x = 88$ riyals.

133 After spending half her money on a book, then 15 riyals on lunch, Sara has 11 riyals left. How much did she start with?

a. 42

b. 52 ✓

c. 62

d. 11

Let the start be x : $\frac{x}{2} - 15 = 11 \Rightarrow \frac{x}{2} = 26 \Rightarrow x = 52$ riyals.

134 After spending half his money on a book, then 21 riyals on lunch, Ali has 17 riyals left. How much did he start with?

- a. 66
- b. 86
- c. 17
- d. 76 ✓

Let the start be x : $\frac{x}{2} - 21 = 17 \Rightarrow \frac{x}{2} = 38 \Rightarrow x = 76$ riyals.

135 After spending half her money on a book, then 13 riyals on lunch, Huda has 18 riyals left. How much did she start with?

- a. 72
- b. 62 ✓
- c. 18
- d. 52

Let the start be x : $\frac{x}{2} - 13 = 18 \Rightarrow \frac{x}{2} = 31 \Rightarrow x = 62$ riyals.

136 After spending half his money on a book, then 19 riyals on lunch, Khalid has 2 riyals left. How much did he start with?

- a. 2
- b. 52
- c. 42 ✓
- d. 32

Let the start be x : $\frac{x}{2} - 19 = 2 \Rightarrow \frac{x}{2} = 21 \Rightarrow x = 42$ riyals.